

# Standard Operating Procedure: BI Publisher Email Bursting Configuration

## 1. Introduction

This document provides the technical implementation details for configuring an automated BI Publisher email bursting solution in Oracle HCM Cloud. The primary objective is to evaluate active contingent worker assignments and trigger automated notifications to line managers regarding contract milestones.

### 1.1 Key Capabilities

- **Automated Daily Trigger:** Evaluates active contingent worker assignments daily.
- **Milestone Alerts:** Triggers notifications when a contract end date is exactly 30, 15, 7, or 1 days away.
- **Latest Contract Enforcement:** Uses a subquery (MAX) to ensure only the latest contract end date is evaluated, suppressing redundant alerts.
- **Individual De-Consolidation:** Employs a unique burst key (BURST\_KEY) for delivering individual inline HTML emails per worker.

## 2. Prerequisites

Before starting the configuration, ensure the following requirements are met:

- Access to BI Publisher (Reports and Analytics) with Developer roles.
- Technical knowledge of Oracle HCM Cloud tables (PER\_ALL\_PEOPLE\_F, PER\_ALL\_ASSIGNMENTS\_M, etc.).
- Authorized outgoing email server configuration for notifications.

## 3. Configuration Steps

### 3.1 Data Model Setup

1. Navigate to **Tools > Reports and Analytics > Browse Catalog**.
2. Click **New > Data Model**.
3. In the left panel, select **Diagram**, click the **+** icon, and select **SQL Query**.
4. Configure properties: Name: **DS\_CONTRACT\_NOTIF**, Source: **ApplicationDB\_HCM**.

5. Paste the following SQL query:

## 3.2 Bursting Engine Setup

1. In the Data Model left pane, click **Bursting** and click **+**.
2. Header Config: Name: **Burst\_To\_Managers**, Split/Deliver By: **/DATA\_DS/G\_1/BURST\_KEY**.
3. Paste the following Bursting SQL:

SQL

```
SELECT
    cwk.full_name AS
WORKER_NAME,
    papf.person_number AS
PERSON_NUMBER,
    paa.employment_category AS
ASSIGNMENT_CATEGORY,
    TO_CHAR(pcf.contract_end_date, 'YYYY-MM-DD') AS
CONTRACT_END_DATE,
    TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE) AS
DAYS_REMAINING,
    pcf.contract_number AS
CONTRACT_NUMBER,
    pcf.type AS
CONTRACT_TYPE,
    CASE
        WHEN TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE) = 30 THEN
'Initial Notification - 30 Days'
        WHEN TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE) = 15 THEN
'Reminder - 15 Days'
        WHEN TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE) = 7 THEN
'Urgent Reminder - 7 Days'
        WHEN TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE) = 1 THEN
'Final Notice - 1 Day'
    END AS
NOTIFICATION_TYPE,
    NVL(mgr_ppn.display_name, 'No Manager Assigned') AS
MANAGER_NAME,
    NVL(mgr_email.email_address, 'No Email Found') AS
MANAGER_EMAIL,
```

```

/* 🔑 UNIQUE BURST KEY PER WORKER */
NVL(mgr_email.email_address, 'No Email Found') || '-' ||
papf.person_number || '-' ||
CASE
    WHEN (TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE)) = 30 THEN
'INITIAL'
    ELSE 'REMINDER'
END
AS BURST_KEY
FROM
    per_cont_workers_current_x          cwk,
    per_all_people_f                    papf,
/* 🔧 Subquery: Evaluates ONLY the latest contract end date per
person */
(
    SELECT p1.person_id, p1.contract_end_date, p1.contract_number,
p1.type
    FROM per_contracts_f p1
    WHERE p1.contract_end_date = (
        SELECT MAX(p2.contract_end_date)
        FROM per_contracts_f p2
        WHERE p2.person_id = p1.person_id
        AND p2.contract_end_date IS NOT NULL
    )
)
    pcf,
per_all_assignments_m                  paa,
(SELECT assignment_id, manager_id, effective_start_date,
effective_end_date FROM per_assignment_supervisors_f WHERE
manager_type = 'LINE_MANAGER') pasf,
(SELECT person_id, display_name, effective_start_date,
effective_end_date FROM per_person_names_f WHERE name_type = 'GLOBAL')
mgr_ppn,
(SELECT person_id, email_address FROM per_email_addresses WHERE
email_type = 'W1') mgr_email
WHERE
    cwk.person_id = papf.person_id
    AND cwk.person_id = pcf.person_id
    AND cwk.person_id = paa.person_id

```

```

AND cwk.assignment_id = paa.assignment_id
AND paa.assignment_id = pasf.assignment_id(+)
AND pasf.manager_id = mgr_ppn.person_id(+)
AND pasf.manager_id = mgr_email.person_id(+)
AND paa.assignment_status_type = 'ACTIVE'
AND paa.employment_category IN ('CATEGORY_1', 'CATEGORY_2',
'CATEGORY_3')
AND (TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE)) IN (30, 15, 7,
1)
AND TRUNC(SYSDATE) BETWEEN papf.effective_start_date AND
papf.effective_end_date
AND TRUNC(SYSDATE) BETWEEN TRUNC(paa.effective_start_date) AND
TRUNC(paa.effective_end_date)
AND TRUNC(SYSDATE) BETWEEN TRUNC(pasf.effective_start_date(+)) AND
TRUNC(pasf.effective_end_date(+))
AND TRUNC(SYSDATE) BETWEEN TRUNC(mgr_ppn.effective_start_date(+))
AND TRUNC(mgr_ppn.effective_end_date(+))
ORDER BY
mgr_email.email_address,
DAYS_REMAINING,
cwk.full_name

```

### 3.3 Report, Layout, and Scheduling

1. Create a Report linked to the Data Model. Upload RTF Layouts: [Initial\\_Layout](#) and [Reminder\\_Layout](#).
2. Schedule via **Scheduled Processes > Schedule New Process** using the daily configuration below.

6. Click **Save Sample Data** and save the Data Model.
- 7.

### 3. Step 2: Bursting Engine Setup

1. In the Data Model left pane, click **Bursting**.
- 2.
3. Click **+** to add a new Bursting definition.
- 4.
5. Configure the Bursting Definition Header:
- 6.

- Name: **Burst\_To\_Managers**
- 
- Type: **SQL Query**
- 
- Data Source: **ApplicationDB\_HCM**
- 
- Split By: **/DATA\_DS/G\_1/BURST\_KEY**
- 
- Deliver By: **/DATA\_DS/G\_1/BURST\_KEY**
- 

7. Paste the following Bursting SQL:

8.

SQL

SQL

S

```
SELECT
mgr_email.email_address || '-' || papf.person_number || '-' ||
CASE
    WHEN (TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE)) = 30 THEN
'INITIAL'
    ELSE 'REMINDER'
END
AS "KEY",
-- Matches BURST_KEY

CASE
    WHEN (TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE)) = 30
    THEN 'Initial_Layout'
    ELSE 'Reminder_Layout'
END
AS "TEMPLATE",
'en-US'
AS "LOCALE",
'HTML'
AS
"OUTPUT_FORMAT",      -- Inline email output
'EMAIL'
AS
"DEL_CHANNEL",
mgr_email.email_address
AS
"PARAMETER1",      -- TO Address
NULL
AS
"PARAMETER2",      -- CC Address
'notifications@company.com'
AS
"PARAMETER3",      -- FROM Address
```

```

CASE
    WHEN (TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE)) = 30
    THEN 'Action Required: Contingent Worker Assignment End (' ||
cwk.full_name || ' )'
    ELSE 'Action Required: Contingent Worker Assignment End -
Reminder (' || cwk.full_name || ' )'
END
"PARAMETER4",          -- Email Subject
NULL
"PARAMETER5",          -- Body Overwrite (N/A for RTF)
'false'
"PARAMETER6",          -- Inline body without PDF attachments
'notifications@company.com'
"PARAMETER7"          -- REPLY-TO Address
FROM
    per_cont_workers_current_x          cwk,
    per_all_people_f                    papf,
    (
        SELECT p1.person_id, p1.contract_end_date, p1.contract_number,
p1.type
        FROM per_contracts_f p1
        WHERE p1.contract_end_date = (
            SELECT MAX(p2.contract_end_date)
            FROM per_contracts_f p2
            WHERE p2.person_id = p1.person_id
            AND p2.contract_end_date IS NOT NULL
        )
    )
    )          pcf,
    per_all_assignments_m                paa,
    (SELECT assignment_id, manager_id, effective_start_date,
effective_end_date FROM per_assignment_supervisors_f WHERE
manager_type = 'LINE_MANAGER') pasf,
    (SELECT person_id, display_name, effective_start_date,
effective_end_date FROM per_person_names_f WHERE name_type = 'GLOBAL')
mgr_ppn,
    (SELECT person_id, email_address FROM per_email_addresses WHERE
email_type = 'W1') mgr_email

```

WHERE

```
    cwk.person_id = papf.person_id
    AND cwk.person_id = pcf.person_id
    AND cwk.person_id = paa.person_id
    AND cwk.assignment_id = paa.assignment_id
    AND paa.assignment_id = pasf.assignment_id(+)
    AND pasf.manager_id = mgr_ppn.person_id(+)
    AND pasf.manager_id = mgr_email.person_id(+)
    AND paa.assignment_status_type = 'ACTIVE'
    AND paa.employment_category IN ('CATEGORY_1', 'CATEGORY_2',
'CATEGORY_3')
    AND (TRUNC(pcf.contract_end_date) - TRUNC(SYSDATE)) IN (30, 15, 7,
1)
    AND TRUNC(SYSDATE) BETWEEN papf.effective_start_date AND
papf.effective_end_date
    AND TRUNC(SYSDATE) BETWEEN TRUNC(paa.effective_start_date) AND
TRUNC(paa.effective_end_date)
    AND TRUNC(SYSDATE) BETWEEN TRUNC(pasf.effective_start_date(+)) AND
TRUNC(pasf.effective_end_date(+))
    AND TRUNC(SYSDATE) BETWEEN TRUNC(mgr_ppn.effective_start_date(+))
AND TRUNC(mgr_ppn.effective_end_date(+))
```

5. Save the Data Model.

6.

## 4. Step 3: Report & Layout Linkage

1. Create a Report linked to the saved Data Model.

2.

3. Upload RTF Layout templates corresponding to the **TEMPLATE** values defined in the Bursting SQL:

4.

- **Layout 1:** **Initial\_Layout**

- 

- **Layout 2:** **Reminder\_Layout**

- 

5. Set default Output Format to **HTML** to support inline rendering.

6.

## 5. Job Execution (Run or Schedule)

Navigate to **Tools > Scheduled Processes > Schedule New Process** and select your BI Publisher Report. Depending on your requirement, proceed with either the Immediate Run or Scheduled Run option.

### Option 1: Immediate Run

1. Select the Report.
2. Click **Submit** to trigger the process immediately.

### Option 2: Scheduled Run

1. Select the Report.
- 1.
2. In the Basic Options, select **Using a Schedule** for the Run Option.
3. Configure the following parameters:
- 4.

| Setting    | Parameter Value                          |
|------------|------------------------------------------|
| Frequency  | Daily                                    |
| Every      | 1 Day(s)                                 |
| Time Zone  | (UTC+00:00) London - GMT (Europe/London) |
| Start Time | 14:00:00 (2:00 PM)                       |
| End Date   | 12/31/4099 (or 'Never')                  |

4. Go to **Advanced Options > Output** and check **Enable Bursting**.
5. Click **Submit**.

## 4. Verification and Monitoring

- **Finding Active Schedules:** Because recurring scheduled jobs do not appear under completed *History*, navigate to **Scheduled Processes > Advanced Search**, set **Status** = **Scheduled / Wait**, clear the submission date, and click **Search**.
- 
- **Bursting Log Auditing:** If emails fail to send, inspect the parent job execution log in BI Publisher under **Manage Report Jobs / Report Job History** to verify key alignment between XML output and the delivery engine.